

From ξ to φ : Recursive Scale Hierarchy

Spectral Gap Structure of FFGFT T^4 and HLV G-Lattice

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Abstract

We study the recursive damping sequence $r(k+1) = r(k)(1 - \xi)$, $r(0) = D_f/3$, which follows solely from the fundamental geometric parameter $\xi = 4/30000$ and the fractal dimension $D_f = 3 - \xi$. At recursion depth $k^* = \log(\varphi)/\xi \approx 3609$ the sequence *passes through* the value $1/\varphi$: $r(3609) \approx 1/\varphi$ with absolute error $4 \cdot 10^{-5}$ (relative $6.5 \cdot 10^{-5}$); the exact real $k_{\text{exact}}^* = 3608.51$ hits $1/\varphi$ identically. **Status clarification (revision 11 June 2026):** $1/\varphi$ is *not* a *fixed point* of the recursion (the unique fixed point is 0) but a *passage point*; since $k^*(c)$ exists for every target $c \in (0, 1)$ with identical parametric status, the ξ^N/φ^N triviality guardrail (Doc. 190 P35) applies to this result in full. It would acquire physical content only through an independent fixing of $k^* \approx 3609$ — currently open. The recursion levels are geometric scale levels with the FFGFT-internal factor $q = 2/3$; $(2/3)^{3609}$ is convergent. The numerical validation study was fully corrected (v4) after an independent audit (P. Stoychev, IPI, 11 June 2026): with a genuine 4D implementation of the T^4 graph and dimension-matched 4D null models, FFGFT T^4 lies *inside* the null band; HLV, with a genuine ensemble, also lies inside the band. Earlier contrary findings (v3) were implementation artefacts and are visibly retracted in this document.

1 Motivation and Context

Doc. 258 showed that the Koide scalar $Q_{\text{FFGFT}} \approx 0.6677$ follows from the geometric invariants r_e, r_μ, r_τ and the lepton exponents p_e, p_μ, p_τ . This value is an *output* of the theory, not an input parameter.

In parallel, the HLV lattice construction of Krüger [1] uses the golden ratio $\varphi = (1 + \sqrt{5})/2$ as the projection ratio in the 6D→3D cut-and-project construction. Krüger's Run E identifies the low-sector gap hierarchy as the only remaining candidate separation channel between HLV and null models.

This document answers three questions:

1. Which scalar is physically correct for the macroscopic geometric structure?
2. How does it connect to ξ ?
3. What does $k^* \approx 3609$ mean physically — and why is it not an astronomical scale?

Remark .1.1 (All three FFGFT scalars are derivable). The question is not which of the three scalars ξ , $1/\varphi$, Q_{FFGFT} *can* be derived from FFGFT — that holds for all three. The question is which one *operates* at which physical level and is therefore responsible for which observables.

2 Fundamental Parameters and Scalar Separation

FFGFT works with the following parameter-free fundamental quantities:

$$\xi = \frac{4}{30000} \approx 1.333 \times 10^{-4} \quad (\text{fractal deviation, microscopic}) \quad (1)$$

$$D_f = 3 - \xi \approx 2.99987 \quad (\text{Hausdorff dimension of } T^4) \quad (2)$$

$$q = \frac{2}{3} \quad (\text{geometric recursion factor}) \quad (3)$$

Definition .2.1 (Three FFGFT scalars near 0.6).

Quantity	Value	Role
$\xi = 4/30000$	1.333×10^{-4}	fundamental input; fractal dimension; active everywhere
$1/\varphi$	0.6180	macroscopic shell ratio; passage point of the ξ -recursion (output; status see Sect. .3)
Q_{FFGFT}	0.6677	Koide scalar leptons (output, Doc. 258); particle physics scale only

All three are derivable from FFGFT. They are *not* interchangeable.

.3 The ξ -Damping Recursion

Definition and Closed Form

Definition .3.1 (ξ -damping recursion). Let $r(0) = D_f/3 = (3 - \xi)/3$ be the shell ratio at the lowest scale level. The recursion reads:

$$r(k+1) = r(k) \cdot (1 - \xi), \quad k = 0, 1, 2, \dots \quad (4)$$

Closed form:

$$r(k) = \frac{D_f}{3} \cdot (1 - \xi)^k \approx \frac{D_f}{3} \cdot e^{-\xi k}. \quad (5)$$

The physical content: $r(0) = D_f/3$ is the ratio of the fractal dimension to the topological dimension 3. Each scale level damps this ratio by the factor $(1 - \xi)$.

Main Observation (Revised): $1/\varphi$ as Passage Point

Remark .3.2 (Revision 11 June 2026). The original version of this section called $1/\varphi$ a *fixed point* of the recursion and stated the result as a theorem. Both were wrong or overstated: the unique fixed point of $r \mapsto r(1 - \xi)$ is $r = 0$; $1/\varphi$ is a *passage point*. The correction follows the audit by P. Stoychev (IPI list, 11 June 2026).

Observation. The sequence $r(k)$ passes through the value $1/\varphi$ at recursion depth

$$k^* = \frac{\log \varphi}{\xi} = \frac{\log\left(\frac{1+\sqrt{5}}{2}\right)}{4/30000} \approx 3609, \quad (6)$$

with $r(3609) = 0.6179940013 \dots$ and $|r(3609) - 1/\varphi| = 4.0 \times 10^{-5}$ (relative 6.5×10^{-5}). The exact real $k_{\text{exact}}^* = (\log(1/\varphi) - \log(D_f/3)) / \log(1 - \xi) = 3608.51$ satisfies $r(k_{\text{exact}}^*) = 1/\varphi$ identically.

Remark .3.3 (Parametric status — the decisive caveat). The arithmetic is correct, but the statement is *not* distinguished: for every target $c \in (0, 1)$ there exists

$$k^*(c) = \frac{\log c - \log(D_f/3)}{\log(1 - \xi)} \quad (7)$$

with exactly the same parametric status — “no free parameter” holds for all targets alike, because the free exponent has merely been renamed k^* and solved for the target:

Target c	$k^*(c)$
$1/\varphi = 0.6180$	3608.5
$Q_{\text{FFGFT}} = 0.6677$	3028.8
$1/2$	5197.9
$1/e$	7499.2
α_{em}	36899.0

The ξ^N/φ^N triviality guardrail (Doc. 190 P35) therefore applies to this very result. The observation would acquire physical content only if $k^* \approx 3609$ were fixed by something independent (a level count of the construction, an observable) and *then* landed on $1/\varphi$. No such fixing is currently known.

.4 Scale Interpretation: $q = 2/3$, Not Factor 2

Remark .4.1 (Critical point). A naive reading might interpret $k^* \approx 3609$ as astronomically many renormalisation-group steps: with factor 2 per step that would be a scale spread of $2^{3609} \approx 10^{1086}$, far beyond any experimental reach.

This objection misses the FFGFT structure.

The four torus lengths follow $L_\mu = L_0 \cdot q^\mu$ with $q = 2/3$. The cumulative scaling after k levels is $S(k) = (2/3)^k \rightarrow 0$.

Proposition .4.2 (Convergence with FFGFT scaling). *Comparison:*

Interpretation	Scale factor at $k^* = 3609$	Character
RG, factor 2 (external convention)	$2^{3609} \approx 10^{1086}$	divergent
FFGFT, $q = 2/3$ (internal)	$(2/3)^{3609} \approx 10^{-635}$	convergent

$k^* \approx 3609$ is a natural number within the theory's scale spectrum, not an astronomical limit.

.5 Spectral Validation Study (Three Layers)

Design

Script version v4 (revised after audit). Two separate sectors are compared, each against dimension-matched null models: *4D sector* — FFGFT T^4 as a kNN graph in genuine 4D coordinates (anisotropic torus lengths $L_\mu = 2\pi(2/3)^\mu$, anisotropy $L_0/L_3 = 3.38$) against three 4D nulls (isotropic cubic, jittered anisotropic, random anisotropic box); *3D sector* — HLV G-Lattice (k NN graph, now with a genuine seed ensemble $\sigma = 0.05$) against three 3D nulls (jittered cubic, random, cubic). $N = 2000$, $k = 16$, three seeds for all models; uniqueness of all point sets is enforced by assertion.

Remark .5.1 (What was wrong in v3 — audit P. Stoychev). (i) The v3 " T^4 graph" used the grid coordinate n_3 only in the acceptance norm, not in the position: of 2000 points only 729 were unique 3D positions; the $1/d^2$ weighting with clamp produced Laplacian eigenvalues up to 2.5×10^{12} . (ii) `hlv_points(seed)` never used its seed — the "ensemble" was the same computation three times. (iii) The shell detector was structurally biased for targets < 1 (see Sect. .5). All three errors are fixed in v4; in addition, 4D models are compared only against 4D nulls, since comparing a 4D graph against 3D nulls would again be unfair.

Layer 1 Gap-hierarchy metrics: max. gap, gap-CV, low-sector $\langle r \rangle$, IDS — analogous to Krüger Run E.

Layer 2 Shell structure: $1/\varphi$ vs. Q_{FFGFT} on the HLV point cloud at $N = 2000$.

Layer 3 Numerical verification of k^* and $r(k^*)$.

Remark .5.2 (Heuristics in the script). The shell-structure function uses a 5% threshold for shell detection. This is a heuristic; for a publication the radial distribution function $g(r)$ would be more robust. The ensemble size (3 seeds) is illustrative, not statistically secured.

Results Layer 1 (v4): Gap Hierarchy ($N = 2000$, 3 Seeds)

Remark .5.3 (Retraction of the v3 result). The result reported in the first version ("FFGFT T^4 graph outside the null band, $\text{gap_CV} = 1.68 \pm 0.09$ ") is **retracted**: it was entirely an artefact of point duplicates (729 unique of 2000 points) combined with the $1/d^2$ clamp. The precision statement "HLV $\approx$ jittered, $\Delta < 0.001$ across 3 seeds" is also withdrawn — it compared a single realisation with itself (seed bug).

Table 1: Gap CV (low sector, $n=100$ modes, $N=2000$, mean $\pm$ std over 3 seeds), v4: separate sectors with dimension-matched null models.

Sector	Model	gap_CV	Band status
4D	FFGFT T^4 (anisotropic)	1.329 ± 0.025	in band
4D	Null: isotropic cubic	2.406 ± 0.010	band edge (top)
4D	Null: jittered anisotropic	0.690 ± 0.026	band edge (bottom)
4D	Null: random anisotropic box	0.740 ± 0.037	–
3D	HLV G-Lattice ($\sigma=0.05$)	0.710 ± 0.008	in band
3D	Null: jittered cubic	0.768 ± 0.011	–
3D	Null: random	0.675 ± 0.043	band edge (bottom)
3D	Null: cubic	0.978 ± 0.010	band edge (top)

Findings:

1. **No model separates.** FFGFT T^4 lies at 1.329 inside the 4D null band $[0.690, 2.406]$; HLV lies at 0.710 inside the 3D null band $[0.675, 0.978]$.
2. **The isotropic 4D lattice itself has the highest gap_CV** (2.406) — regular lattices produce strong eigenvalue degeneracies in 4D. A high gap_CV is therefore not a T^4 signature but a generic order signature.
3. **Robustness checked:** the T^4 result is jitter-stable ($\sigma = 0.02\text{--}0.10$: 1.43/1.35/1.35) and consistent for $n_{\text{range}} = 4/5$ (1.35/1.40); $n_{\text{range}} = 3$ (2.28) is a small-size artefact.
4. **HLV with a genuine ensemble** remains in the band and close to jittered cubic — the v3 *finding* (consistent with Krüger's Run E) survives, the precision claimed then does not.

Remark .5.4 (Consequence for the comparison). At $N = 2000$ the gap_CV diagnostic separates neither the T^4 graph nor HLV from dimension-matched null models. This *reinforces* Krüger's conservative Run-E finding from an independent pipeline and further constrains the common-spectral-signature hypothesis of Doc. 271: at this size, no signature is demonstrable at the graph level.

Results Layer 2: Shell Structure — Retracted

Remark .5.5 (Retraction of the v3 discrimination). The discrimination reported in the first version ($\text{err}(Q_{\text{FFGFT}}) < \text{err}(1/\varphi)$, $\Delta = 0.145$) is **retracted**. The detector identifies shells as a

strictly increasing distance sequence (factor ≥ 1.05); for any target ratio $c < 1$ the minimum achievable error is $(1.05/c - 1)$: 0.699 for $1/\varphi$, 0.573 for Q_{FFGFT} . *The larger of two sub-1 ratios wins on any point set whatsoever* — the script's own controls showed this (cubic: 0.976 vs. 1.135; random: 1.194 vs. 1.371) but were not read as an alarm. The finding was structural, not physical (audit P. Stoychev).

In v4 the detector raises an exception for targets ≤ 1 ; only ratios > 1 are tested. Reference value: $\text{err}(\varphi=1.618)_{\text{HLV}} = 0.257$; the best-fit consecutive shell ratio of the HLV point cloud is, per Stoychev's measurement, ≈ 1.15 — neither φ , $1/\varphi$, nor Q_{FFGFT} . For future versions, the radial distribution function $g(r)$ with peak-position analysis is the more robust route.

Result Layer 3: Numerical Verification

$$k^* = \frac{\log \varphi}{\xi} \approx 3609.1, \quad r(3609) = 0.6179940013 \dots \approx \frac{1}{\varphi} \quad (|\text{error}| = 4.0 \times 10^{-5}, \text{ relative } 6.5 \times 10^{-5}). \quad (8)$$

The arithmetic is correct; the error statement was made precise in v3 (previously incorrectly $< 10^{-9}$). For the *status* of the statement — passage point, not fixed point; $k^*(c)$ exists for every target — see Sect. .3: “without free parameters” does not distinguish this result.

.6 Discussion

Relation to the φ^N -triviality warning (Doc. 190 P35) — Revised

Remark .6.1 (Retraction of the demarcation). The first version of this section gave three arguments why the triviality warning would *not* apply to the k^* result (“no free exponent”, “no empirical anchor”, “no external calibration”). This demarcation is **retracted**: all three arguments overlook that the free exponent has merely been *renamed* k^* and *solved for* the desired target. $k^*(c)$ exists for every $c \in (0, 1)$ with identical properties (table in Sect. .3); the selection of the target $1/\varphi$ is the hidden choice. P35 applies in full — this is exactly the case the guardrail was written for, and applying it to one's own result is its first serious corpus-internal stress test (passed: the result is downgraded).

What remains: the three-level separation of φ -roles documented in Doc. 190 P37 — (1) microscopic/ ξ -level, (2) macroscopic/ $1/\varphi$ as passage point of the recursion, (3) particle physics/ Q_{FFGFT} — remains valid and useful as a *conceptual separation*. Doc. 188 uses φ^{-1} as a phenomenological scaling factor between fractal levels; whether this factor follows structurally from the ξ -recursion is, after the present revision, again an *open question*: the recursion delivers it only if the depth $k^* \approx 3609$ is independently justified.

Position in the Document Series

Doc. 270 describes the reduction chain $T^4 \rightarrow T^0$ as two qualitatively different operation types. The present document adds the *scale hierarchy*: the $q = 2/3$ recursion connects the microscopic ξ -level with the macroscopic φ -level through cumulative geometric damping — without a qualitative change in operation type.

Doc. 271 provides a structural comparison of FFGFT and HLV. The present document makes the spectral comparison announced there concrete through the numerical script and identifies `gap_CV` as a shared diagnostic channel.

Open Points

1. Shell diagnostics on a $g(r)$ basis (peak positions, ratios > 1) — the v3 detector is unusable for sub-1 targets (Sect. .5).
2. Pre-declared ensemble band tests analogous to Krüger Run E (metrics, bands and seeds fixed before the computation) for the 4D T^4 graph.
3. Larger N and more seeds: the present 3-seed ensembles are illustrative, not statistically secured.
4. An independent fixing of $k^* \approx 3609$ (a level count of the construction or a physical observable) — without it the passage-point observation has no physical content (Sect. .3). Currently not known.
5. Cross-validation of the v4 implementation against the deduplicated/seeded variants offered by P. Stoychev.

.7 Conclusion

1. The ξ -damping recursion passes through $1/\varphi$ at $k^* = \log(\varphi)/\xi \approx 3609$; the arithmetic is correct (error $4 \cdot 10^{-5}$), but $1/\varphi$ is a *passage point*, not a *fixed point*, and $k^*(c)$ exists for every target — the P35 triviality guardrail applies; without an independent fixing of k^* the observation has no physical content.
2. With $q = 2/3$, $(2/3)^{3609}$ is convergent and internally consistent, not an astronomical limit — this scale interpretation stands.
3. **v4 main result ($N = 2000$, dimension-matched nulls):** No separation. FFGFT T^4 (genuine 4D, gap_CV = 1.329 ± 0.025) lies inside the 4D null band $[0.690, 2.406]$; HLV (genuine ensemble, 0.710 ± 0.008) lies inside the 3D null band $[0.675, 0.978]$. This reinforces Krüger's conservative Run-E finding from an independent pipeline.
4. The v3 findings " T^4 outside the band" and " Q_{FFGFT} discriminates against $1/\varphi$ " are **retracted** (duplicate artefact and detector bias, respectively); the HLV ensemble seed bug is fixed. All three errors trace back to the audit by P. Stoychev and are registered as K4 in Doc. 190.
5. The three scalars ξ , $1/\varphi$, Q_{FFGFT} remain useful as a conceptual separation at different physical levels (Doc. 190 P37); the *derivation claim* for $1/\varphi$ from the recursion is downgraded to an open question (Doc. 190 P38).

Connections. Doc. 258 (Q_{FFGFT} , Koide ladder), Doc. 270 (reduction chain $T^4 \rightarrow T^0$), Doc. 271 (FFGFT-HLV structural comparison), Doc. 272 (HLV/Dot-Theory context). The Python script `ffgft_hlv_gap_v4.py` (corrected version; v2/v3 remain in the archive as documented error states) is publicly available and reproducible.

Acknowledgement. Plamen Stoychev (IPI) for the independent audit of 11 June 2026, which identified all three script errors and forced the status clarification of the k^* result. The revision of this document implements his audit in full.

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